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TREND-BASED MOCK PAPER

GATE Biotechnology

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Biotechnology
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Choose similar to 'catalyst':

(Numerical Answer Type — Enter the numerical value)

Section B: Biotechnology

Q2. [1-NAT] [1 mark]

Normal distribution mean = median = ?

(Numerical Answer Type — Enter the numerical value)

Q3. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
 - (B) 2
 - (C) 16
 - (D) 1
-

Q4. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
 - (B) 0.092
 - (C) 0.762
 - (D) 0.195
-

Q5. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2

- (B) 0.5
- (C) 0.6
- (D) 0.4

Q6. [1-MCQ] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q7. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q8. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q9. [2-NAT] [2 marks]

The Laplace transform of $t * e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q10. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) $\pi * r^2$
- (B) $2 * \pi * r$
- (C) $\pi * r$
- (D) $2 * \pi * r^2$

Q11. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)} * u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q12. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q13. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q14. [2-MSQ] [2 marks]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q15. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q16. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q17. [2-NAT] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\operatorname{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\operatorname{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)(1+z)}$

Q19. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q20. [2-MSQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16

(D) 12

Q21. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q22. [2-NAT] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-NAT] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q25. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q26. [2-MSQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q27. [2-NAT] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q28. [2-MSQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q29. [2-MCQ] [2 marks]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q30. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q31. [2-NAT] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q32. [1-MCQ] [1 mark]

Central dogma: reverse transcription is:

- (A) RNA $\rightarrow$ DNA
- (B) DNA $\rightarrow$ RNA
- (C) Protein $\rightarrow$ DNA
- (D) RNA $\rightarrow$ Protein

Q33. [1-MCQ] [1 mark]

Photosynthesis light reactions produce:

- (A) Glucose only
- (B) ATP and NADPH
- (C) CO₂
- (D) O₂ only

Q34. [1-MCQ] [1 mark]

Competitive enzyme inhibition: K_m

- (A) Increases
- (B) Decreases
- (C) Unchanged
- (D) Zero

Q35. [1-NAT] [1 mark]

Mendelian dihybrid cross (both heterozygous) ratio:

(Numerical Answer Type — Enter the numerical value)

Q36. [1-MCQ] [1 mark]

DNA replication in eukaryotes occurs in:

- (A) Cytoplasm
- (B) Nucleus
- (C) Mitochondria only
- (D) G₁ phase

Q37. [2-MCQ] [2 marks]

Photosynthesis light reactions produce:

- (A) Glucose only
- (B) ATP and NADPH
- (C) CO₂
- (D) O₂ only

Q38. [2-MSQ] [2 marks]

Mendelian dihybrid cross (both heterozygous) ratio:

- (A) 3:1
- (B) 9:3:3:1
- (C) 1:2:1
- (D) 1:1:1:1

Q39. [2-NAT] [2 marks]

DNA replication in eukaryotes occurs in:

(Numerical Answer Type — Enter the numerical value)

Q40. [2-NAT] [2 marks]

Photosynthesis light reactions produce:

(Numerical Answer Type — Enter the numerical value)

Q41. [2-NAT] [2 marks]

Mendelian dihybrid cross (both heterozygous) ratio:

(Numerical Answer Type — Enter the numerical value)

Q42. [2-NAT] [2 marks]

Mendelian dihybrid cross (both heterozygous) ratio:

(Numerical Answer Type — Enter the numerical value)

Q43. [1-NAT] [1 mark]

DNA stands for:

(Numerical Answer Type — Enter the numerical value)

Q44. [1-MCQ] [1 mark]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA

Q45. [1-MCQ] [1 mark]

Linked genes show:

- (A) Independent assortment
- (B) No recombination
- (C) Less than 50% recombination
- (D) Always 50%

Q46. [1-MCQ] [1 mark]

mRNA processing:

- (A) Transcription only
- (B) Capping, splicing, poly-A
- (C) Translation only
- (D) Replication

Q47. [1-NAT] [1 mark]

DNA B-form base pairs per turn:

(Numerical Answer Type — Enter the numerical value)

Q48. [1-MCQ] [1 mark]

AaBb x aabb. Genotypic ratio:

- (A) 1:1:1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:2:1

Q49. [1-NAT] [1 mark]

ABO blood group has how many alleles?

(Numerical Answer Type — Enter the numerical value)

Q50. [2-MCQ] [2 marks]

DNA B-form base pairs per turn:

- (A) 10
- (B) 10.4
- (C) 12
- (D) 8

Q51. [2-NAT] [2 marks]

Linked genes show:

(Numerical Answer Type — Enter the numerical value)

Q52. [1-MCQ] [1 mark]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1
- (D) -1

Q53. [1-NAT] [1 mark]

Strongest acid:

(Numerical Answer Type — Enter the numerical value)

Q54. [1-MCQ] [1 mark]

First-order half-life with $k = 0.693 \text{ min}^{-1}$:

- (A) 1 min
- (B) 0.693 min
- (C) 1.44 min
- (D) 0.5 min

Q55. [1-MCQ] [1 mark]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8
- (D) 6.98

Q56. [1-MCQ] [1 mark]

Galvanic cell converts:

- (A) Electrical to chemical
- (B) Chemical to electrical
- (C) Heat to work
- (D) Mechanical to electrical

Q57. [2-MCQ] [2 marks]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8
- (D) 6.98

Q58. [2-MCQ] [2 marks]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8
- (D) 6.98

Q59. [2-MCQ] [2 marks]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1
- (D) -1

Q60. [1-MCQ] [1 mark]

Amino acids linked by:

- (A) Peptide bond
- (B) Hydrogen bond
- (C) Ionic bond
- (D) None

Q61. [1-MCQ] [1 mark]

Binary fission in:

- (A) Bacteria
- (B) Fungi
- (C) Plants
- (D) Animals

Q62. [1-MCQ] [1 mark]

Batch process is:

- (A) Continuous
- (B) Intermittent
- (C) Steady
- (D) None

Q63. [1-MCQ] [1 mark]

Antibody produced by:

- (A) T cells
- (B) B cells
- (C) Macrophages
- (D) None

Q64. [1-NAT] [1 mark]

BLAST is for:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Chromosome number in daughter:

- (A) Halved
 - (B) Same
 - (C) Doubled
 - (D) Variable
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1NAT	C	1
3	1NAT	A	1
4	1MCQ	A	1
5	1MCQ	A	1
6	1MCQ	A	1
7	1MCQ	B	1
8	1MCQ	B	1
9	1NAT	A	1
10	1MCQ	B	1
11	1MCQ	B	1
12	1MCQ	A	1
13	1MCQ	B	1

14	1MCQ	C	1
15	1MCQ	A	1
16	1NAT	D	1
17	1NAT	B	1
18	1MCQ	B	1
19	1MCQ	A	1
20	1NAT	B	1
21	1MCQ	A	1
22	1MCQ	D	1
23	1MCQ	B	1
24	1NAT	B	1
25	1MCQ	A	1
26	1MCQ	A	1
27	1MCQ	B	1
28	1MCQ	B	1

29	1MCQ	A	1
30	1MCQ	C	1
31	2MCQ	B	2
32	2NAT	A	2
33	2MCQ	A	2
34	2MCQ	A	2
35	2MCQ	A	2
36	2MCQ	A	2
37	2MSQ	C	2
38	2MCQ	C	2
39	2MCQ	B	2
40	2NAT	A	2
41	2MCQ	A	2
42	2MCQ	A	2
43	2MSQ	C	2

44	2MCQ	A	2
45	2NAT	A	2
46	2NAT	B	2
47	2MCQ	A	2
48	2MCQ	C	2
49	2MSQ	B	2
50	2NAT	B	2
51	2MSQ	A	2
52	2MCQ	D	2
53	2MCQ	C	2
54	2MCQ	B	2
55	2MSQ	B	2
56	2NAT	B	2
57	2NAT	B	2
58	2MCQ	A	2

59	2NAT	B	2
60	2MCQ	B	2
61	2MCQ	D	2
62	2NAT	C	2
63	2NAT	A	2
64	2MCQ	B	2
65	2NAT	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: difficult

Q2.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Probability | Difficulty: moderate

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Genetics | Topic: Molecular Biology | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Genetics | Topic: Mendel | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Biochemistry | Topic: Proteins | Difficulty: easy

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Microbiology | Topic: Bacteria | Difficulty: difficult

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Bioprocess Engineering | Topic: Fermentation | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Immunology | Topic: Antibodies | Difficulty: difficult

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Bioinformatics | Topic: Sequence | Difficulty: moderate

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Cell Biology | Topic: Mitosis | Difficulty: difficult

Q11.

Answer: B

Solution:

$\Delta n = 2 - 4 = -2$.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q12.

Answer: A

Solution:

Reverse transcription: RNA→DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q13.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q14.

Answer: C

Solution:

Linked genes on same chromosome: recombination < 50%.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q15.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q16.

Answer: D

Solution:

Perchloric acid (HClO₄) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q17.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: easy

Q18.

Answer: B

Solution:

Pre-mRNA: 5 cap, intron splicing, 3 poly-A tail.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q19.

Answer: A

Solution:

$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min.}$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q20.

Answer: B

Solution:

B-DNA: ~10.4 bp per turn.

Subject: Genetics | Topic: Genetics | Difficulty: easy

Q21.

Answer: A

Solution:

AaBb x aabb: offspring AaBb, Aabb, aaBb, aabb in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q22.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{\text{total}}$ approx $1.1e-7$. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q23.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q24.

Answer: B

Solution:

ABO: I^A , I^B , i (three alleles).

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q25.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q26.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: difficult

Q28.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q29.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q31.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q32.

Answer: A

Solution:

$L\{f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q33.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q35.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q39.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q44.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: B

Solution:

$\text{proj} = ((v.w)/(w.w))*w = (4/5)*(2,1,0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: A

Solution:

Area = $\pi*r^2$. Circumference = $2*\pi*r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q52.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{\text{total}}$ approx $1.1e-7$. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q53.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q55.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q56.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q57.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q58.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q59.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q60.

Answer: B

Solution:

B-DNA: ~10.4 bp per turn.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q61.

Answer: D

Solution:

Very dilute: [H⁺] from water dominates. [H⁺]_{total} approx 1.1e-7. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q62.

Answer: C

Solution:

Linked genes on same chromosome: recombination < 50%.

Subject: Genetics | Topic: Genetics | Difficulty: easy

Q63.

Answer: A

Solution:

$L\{f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: B

Solution:

$\Delta_n = 2 - 4 = -2$.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q65.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 30 questions, 54 marks
Biology: 11 questions, 17 marks
Genetics: 9 questions, 11 marks
Chemistry: 8 questions, 11 marks
General Aptitude: 1 questions, 1 marks
Biochemistry: 1 questions, 1 marks
Microbiology: 1 questions, 1 marks
Bioprocess Engineering: 1 questions, 1 marks
Immunology: 1 questions, 1 marks
Bioinformatics: 1 questions, 1 marks
Cell Biology: 1 questions, 1 marks

Difficulty Distribution:

Easy: 16 | Moderate: 40 | Difficult: 9

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.